

We'll assume the file:

names.txt

contains:

Aarav Sharma

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Priya Singh

Ananya Gupta

cut Command – COMPLETE NOTES

◆ **What is cut?**

cut is a **text filter** used to **extract specific characters, bytes, or fields** from each line of a file or input.

It works **line by line**.

◆ **General Syntax**

cut OPTION [RANGE] [FILE]

Or with pipe:

command | cut OPTION [RANGE]

1 Character-based Cutting (-c)

Used when you want **specific character positions**.

◆ **Single character**

cut -c 1 names.txt

👉 Prints **1st character** of each line.

◆ **Multiple characters**

cut -c 1,3,5 names.txt

👉 Prints **1st, 3rd, and 5th characters**.

◆ **Character range**

cut -c 1-5 names.txt

👉 Prints characters **1 to 5**.

◆ **Open-ended range**

cut -c 5- names.txt

👉 From **5th character to end**.

cut -c -4 names.txt

👉 From **1st to 4th character**.

◆ **Combined ranges**

cut -c 1-3,6-8 names.txt

2 Byte-based Cutting (-b)

Works like -c, but counts **bytes instead of characters** (important for multibyte characters).

◆ Single byte

```
cut -b 1 names.txt
```

◆ Byte range

```
cut -b 1-5 names.txt
```

◆ Open range

```
cut -b 3- names.txt
```

📌 Difference from -c:

- -b → bytes
- -c → characters

3 Field-based Cutting (-f)

Used when data is **column-based** and separated by a **delimiter**.

◆ Mandatory option: delimiter (-d)

◆ Default delimiter

- TAB (if -d not specified)

◆ Extract first field

```
cut -d " " -f 1 names.txt
```

👉 First names only.

◆ Extract multiple fields

```
cut -d " " -f 1,2 names.txt
```

◆ Field range

```
cut -d " " -f 1-2 names.txt
```

◆ Open-ended fields

```
cut -d " " -f 2- names.txt
```

👉 Field 2 to end.

```
cut -d " " -f -1 names.txt
```

👉 Field 1 only.

◆ Using other delimiters

Colon (:)

```
cut -d ":" -f 1 /etc/passwd
```

Comma (,)

```
cut -d "," -f 2 data.csv
```

4 Suppress Lines Without Delimiter (-s)

By default, cut prints **entire line** if delimiter is missing.

◆ Without -s

```
cut -d ":" -f 1 file.txt
```

◆ With -s

```
cut -d ":" -f 1 -s file.txt
```

👉 Lines **without delimiter** are skipped

5 Output Delimiter (--output-delimiter)

Changes the delimiter in output.

```
cut -d " " -f 1,2 --output-delimiter="," names.txt
```

Output:

Aarav,Sharma

Rohan,Verma